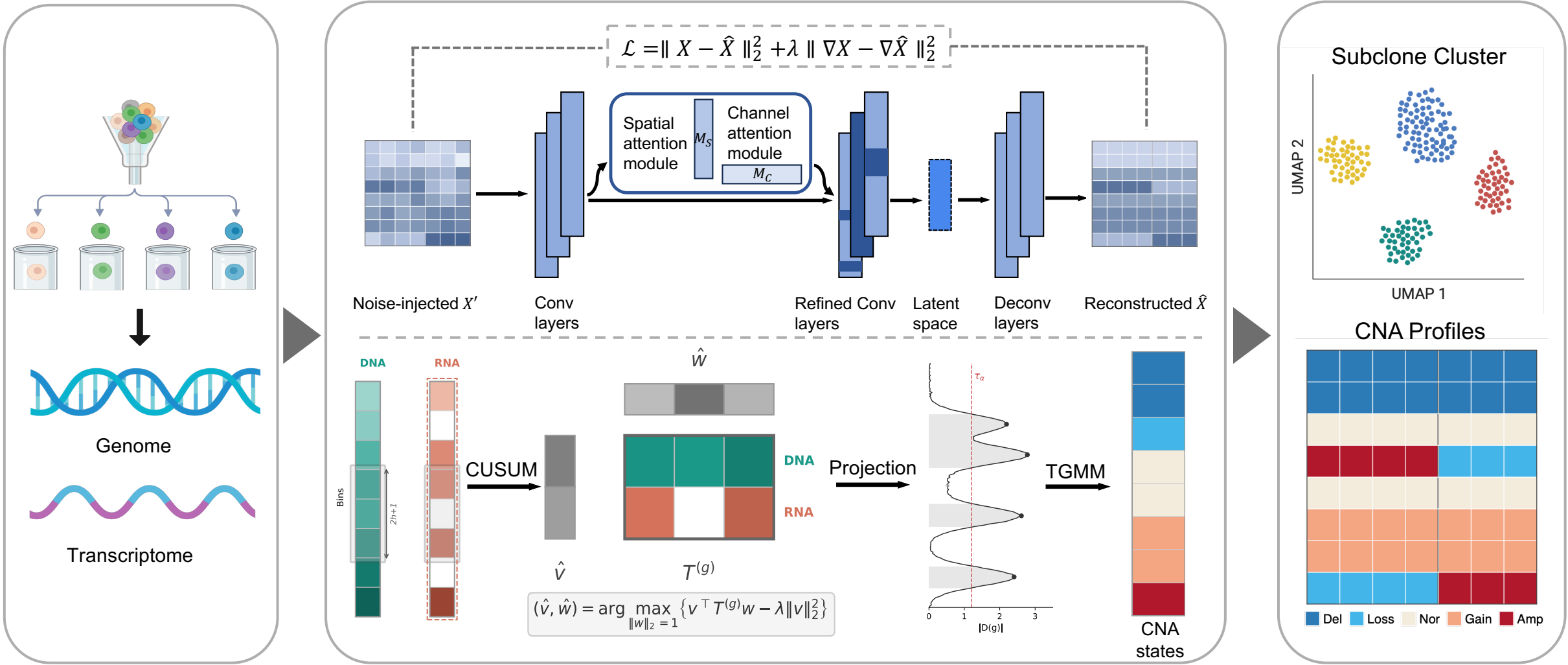

Genome
Transcriptome
$\mathcal{L} = \| X - \hat{X} \|_2^2 + \lambda \| \nabla X - \nabla \hat{X} \|_2^2$
Spatial attention module
M_S
Channel attention module
M_C
Noise-injected X'
Conv layers
Refined Conv layers
Latent space
Deconv layers
Reconstructed $\hat{X}$
DNA
RNA
Bins
$2h+1$
CUSUM
$\hat{v}$
$\hat{W}$
DNA
RNA
$T^{(g)}$
$(\hat{v}, \hat{w}) = \arg\max_{\|w\|_2 = 1} \{ v^\top T^{(g)} w - \lambda \|v\|_2^2 \}$
Projection
τ_α
$|D(g)|$
TGMM
CNA states
Subclone Cluster
UMAP 2
UMAP 1
CNA Profiles
Del
Loss
Nor
Gain
Amp